

WEEK 4 PROGRESS UPDATE

Multilingual Speech-to-Intent Detection System

Research Progress Report

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Reporting Period	Week 4	Status	Ongoing — Model Training & Evaluation Phase

1. Project Objective

The objective of the current work is to build a multilingual speech-based intent detection pipeline for Indian languages.

The planned pipeline is:

Speech Audio → Whisper Speech-to-Text → Transcribed Text → MuRIL Intent Classifier → Predicted Intent

The current experiments focus on five Indian languages:

- Hindi
- Kannada
- Malayalam
- Tamil
- Telugu

The intent classification task contains 60 different intent categories.

2. Initial Whisper Experiment

The first stage of the pipeline was focused on the speech-to-text component using OpenAI Whisper.

Whisper Small

Initially, the Whisper Small model was loaded and tested with an Indian-language audio sample. The purpose was to check whether Whisper could correctly recognize the spoken language, convert speech into text, and produce usable text for the downstream intent classifier.

The initial transcription showed errors in recognizing the spoken Indian-language audio.

Whisper Medium

To investigate whether a larger Whisper model would improve transcription quality, the Whisper Medium model was loaded and tested on the same type of audio.

The Medium model produced better transcription behaviour compared with the Small model in our testing, which motivated the decision to use the Medium model for the current pipeline. The current Whisper model is loaded on the available CUDA GPU.

3. PyTorch and GPU Setup

PyTorch was used as the main deep-learning framework. It is responsible for the tensor-based mathematical operations required by the neural networks and also allows the models to run on the GPU.

The Colab environment was verified to have:

- PyTorch: 2.11.0+cu128
- CUDA: Available
- GPU: NVIDIA Tesla T4
- GPU memory: approximately 14.56 GB

The GPU was used for both efficient model loading and MuRIL fine-tuning.

4. MASSIVE Dataset Preparation

The MASSIVE multilingual intent dataset was prepared for the intent-classification stage.

The original dataset contained several fields. For the MuRIL classification task, the relevant information was reduced to:

- Text/utterance
- Intent label

The utterance was stored as text and the corresponding intent was stored as labels. The five selected Indian-language datasets were combined into one multilingual dataset.

The resulting dataset contained:

- Training examples: 57,570
- Validation examples: 10,165
- Test examples: 14,870
- Languages: 5
- Intent categories: 60

The test set was kept separate and was used later as an unseen evaluation set.

5. MuRIL Tokenization

The multilingual text was passed through the MuRIL tokenizer.

The purpose of tokenization was to convert human-readable Indian-language text into numerical token IDs that the MuRIL neural network can process.

The tokenizer generated representations such as:

- input_ids
- token_type_ids
- attention_mask

A maximum sequence length of 128 was used, with truncation enabled. The dataset was tokenized successfully across all three splits:

- Training: 57,570 examples
- Validation: 10,165 examples
- Test: 14,870 examples

6. MuRIL Intent Classifier

The pretrained google/muril-base-cased model was loaded.

Instead of using MuRIL only as a general pretrained language model, a 60-class intent classification head was added on top of the pretrained MuRIL encoder. The classifier therefore learns to map an input sentence to one of the 60 intent categories.

The MuRIL model has a hidden size of 768.

The classification layer was newly initialized because the original pretrained MuRIL model was not specifically trained for our 60 MASSIVE intent categories.

7. Evaluation Metrics

Three evaluation metrics were implemented:

Accuracy

Measures the percentage of predictions that are completely correct.

Macro F1

Calculates F1 separately for each intent and then gives equal importance to every intent. This is especially important because some intents have fewer examples than others.

Weighted F1

Calculates F1 while taking the number of examples in each intent into account.

Therefore, all three metrics were used rather than relying only on accuracy.

8. Dynamic Data Collator and Batch Processing

A dynamic data collator was implemented using `DataCollatorWithPadding`. Its purpose is to dynamically pad sentences within each training batch so that sentences with different lengths can be processed efficiently.

The training batch size was set to 16. The model therefore receives the training data in batches rather than processing all 57,570 examples simultaneously. This makes training possible within the available GPU memory and allows efficient GPU computation.

9. Initial MuRIL Training — 2 Epochs

The first major MuRIL fine-tuning experiment used 2 epochs. The configuration included:

- Training examples: 57,570
- Validation examples: 10,165
- Test examples: 14,870
- Languages: 5
- Intents: 60
- Batch size: 16
- Learning rate: 2×10^{-5}
- Weight decay: 0.01
- Warmup steps: 0
- FP16 mixed precision: enabled
- Epochs: 2

2-Epoch Results

The final unseen test results were:

- Accuracy: 72.70%
- Macro F1: 46.75%
- Weighted F1: 68.06%

The result showed that MuRIL was able to learn the multilingual intent-classification task, but there was significant room for improvement.

10. Per-Language Performance After 2 Epochs

The performance was relatively similar across the five languages.

Language	Accuracy	Macro F1	Weighted F1
Hindi	73.64%	46.97%	68.86%
Kannada	72.53%	46.41%	67.84%
Malayalam	72.49%	46.85%	67.85%
Tamil	72.60%	46.80%	68.00%
Telugu	72.23%	46.71%	67.77%

This showed that the model did not completely fail on any particular language, although overall intent classification performance was still limited.

11. Intent-Level Error Analysis

After the initial training, the performance of individual intents was analyzed.

Some intents performed extremely well, including:

- weather_query
- transport_traffic
- calendar_remove
- qa_currency
- calendar_set
- email_query
- email_sendemail
- datetime_query
- play_music
- lists_remove

However, several intents had very poor performance, including some with $F1 = 0$. Examples included:

- takeaway_query
- general_greet
- music_dislikeliness
- iot_wemo_off
- email_addcontact
- audio_volume_other
- play_audiobook
- iot_hue_lightup
- iot_wemo_on
- general_joke

This showed that the overall accuracy did not represent all intents equally well.

12. Common Intent Confusions

The confusion analysis showed that the model was frequently confusing semantically similar intents.

Some important examples were:

- iot_hue_lightchange → iot_hue_lightoff
- play_audiobook → play_podcasts

- play_game → play_radio
- iot_hue_lightup → iot_hue_lightoff
- music_likeness → music_query
- iot_cleaning → audio_volume_mute
- social_query → social_post
- transport_taxi → transport_ticket
- alarm_query → alarm_set
- takeaway_query → recommendation_locations

This analysis helped identify where the classifier was actually struggling.

13. End-to-End Speech-to-Intent Pipeline Test

After training the text-based intent classifier, the complete pipeline was also tested. The process was:

Audio → Whisper → Transcribed Text → MuRIL Tokenizer → MuRIL Classifier → Intent Prediction

A test audio file was uploaded and passed to Whisper. Whisper automatically detected the language and produced a transcription. The transcription was then passed directly to the trained MuRIL classifier. This confirmed that the separate components can be connected into an end-to-end pipeline.

However, the end-to-end test also revealed an important issue: Whisper's automatic language detection and transcription can sometimes be incorrect for short or difficult Indian-language speech. Therefore, the quality of the speech-to-text stage directly affects the downstream intent classification stage.

14. Major Experiment — Increasing Training Epochs

The next experiment was performed by changing the number of training epochs. The original experiment used 2 epochs. This was changed to 5 epochs. The other major training settings were kept the same so that the effect of increasing the training duration could be studied.

The 5-epoch training produced the following validation results:

Epoch	Train Loss	Val Loss	Accuracy	Macro F1	Weighted F1
1	1.5266	1.5093	64.55%	35.19%	58.66%
2	0.7543	0.8974	81.66%	66.75%	80.37%
3	0.4082	0.7852	83.49%	73.04%	82.69%
4	0.2878	0.7842	85.44%	78.95%	85.15%
5	0.1690	0.8025	85.99%	80.29%	85.67%

The model continued improving in accuracy and F1 across the five epochs.

15. Final Unseen Test Results After 5 Epochs

The 5-epoch model was evaluated on the completely unseen test set. Results:

- Test Accuracy: 86.40%
- Test Macro F1: 79.18%
- Test Weighted F1: 86.18%
- Test loss: 0.7760

This was a major improvement over the 2-epoch experiment.

Comparison

Metric	2 Epochs	5 Epochs	Improvement
Accuracy	72.70%	86.40%	+13.70 pp
Macro F1	46.75%	79.18%	+32.43 pp
Weighted F1	68.06%	86.18%	+18.12 pp

The large improvement in Macro F1 is particularly important because it indicates that performance improved substantially across the different intent categories rather than only improving on the most frequent classes.

16. Per-Language Performance After 5 Epochs

The 5-epoch model also showed strong and relatively balanced performance across all five languages.

Language	Accuracy	Macro F1	Weighted F1
Hindi	87.69%	80.65%	87.41%
Kannada	85.71%	78.48%	85.43%
Malayalam	86.68%	79.63%	86.51%
Tamil	85.81%	78.06%	85.61%
Telugu	86.11%	79.12%	85.97%

The performance difference between the highest and lowest language accuracy was relatively small, indicating reasonably balanced performance across the five languages.

17. Current Findings

The experiments conducted so far provide several important findings:

- Whisper can be used as the speech-to-text component of the pipeline.
- Whisper Medium was tested after the initial Small model to investigate whether a larger ASR model would improve transcription.

- MuRIL can successfully be fine-tuned for multilingual Indian-language intent classification.
- The combined dataset of five Indian languages can be used to train a single multilingual intent classifier.
- The initial 2-epoch training was not sufficient to obtain the best performance.
- Increasing training from 2 to 5 epochs produced a substantial improvement.
- The test accuracy increased from 72.70% to 86.40%.
- Macro F1 increased from 46.75% to 79.18%.
- Performance became much more balanced across the five languages.
- Some individual intents still remain difficult and require further investigation.
- Semantically similar intents are still a major source of classification errors.
- The end-to-end pipeline works, but Whisper transcription quality can affect the final intent prediction.

18. Current Status of the Project

The current pipeline has reached the following stage:

Speech Audio → Whisper Speech-to-Text → Indian-language Transcription → MuRIL Tokenizer → MuRIL Base Model → 60-Class Intent Classifier → Predicted Intent

The current best text-classification result is:

86.40% accuracy, 79.18% Macro F1, and 86.18% Weighted F1 on the unseen test set.

Therefore, the basic multilingual intent-classification pipeline is functioning successfully, and the current phase is focused on systematic experimentation and improvement rather than building the pipeline from scratch.

19. Overall Progress

The project has progressed from an initial speech-to-text experiment to a complete multilingual speech-to-intent pipeline.

The most important progress so far is the improvement from:

- 72.70% test accuracy with 2 epochs
- to: 86.40% test accuracy with 5 epochs

This confirms that the current MuRIL architecture and dataset are capable of learning the 60-intent multilingual classification task effectively, while also showing that further controlled experiments are required to identify the remaining limitations and improve the system further.